Unified Detection Framework for Robbery Events: Integrating YOLOv8, Fast R-CNN, and RetinaNet with Explainable AI Validation and Real-time Android Application Integration

***Abstract*—This research proposes a Unified Detection Framework for Robbery Events by combining YOLOv8, Fast R-CNN, and RetinaNet models. The framework incorporates Explainable AI Validation for transparency and real-time detection capabilities. Additionally, an Android application integration is implemented for practical deployment. Through experimentation and evaluation, the effectiveness of the framework in detecting robbery events is demonstrated. The fusion of multiple object detection models enhances accuracy and efficiency, while the integration of explainable AI validation ensures interpretability of results. The proposed framework shows promising results for addressing security concerns in real-world scenarios.**

***Index Terms*—Robbery Detection, Multi-Model Deep Learning Fusion, YOLOv8n, Faster RCNN, RetinaNet**

I. Introduction

In today's rapidly evolving technological landscape, there is an increasing demand for robust and efficient surveillance systems to combat criminal activities. In response to this need, this research presents a pioneering Unified Detection Framework for Robbery Events. In contrast to traditional approaches that often rely on single-object detection models, our framework integrates the capabilities of three state-of-the-art models: YOLOv8, Fast R-CNN, and RetinaNet. By combining these models, our goal is not only to improve the accuracy of identifying robbery events but also to significantly enhance the efficiency of detection in real-time scenarios. Additionally, the incorporation of Explainable AI Validation mechanisms ensures transparency and interpretability of the detection process, which is essential for establishing trust and promoting collaboration between humans and machines in security applications. Moreover, the practicality of our framework is emphasized by its seamless integration into an Android application, enabling swift deployment on mobile platforms. Through thorough experimentation and evaluation, we demonstrate the effectiveness of our proposed framework in detecting robbery events, offering promising prospects for addressing security concerns in various real-world settings. The contemporary surveillance systems are confronted with challenges in effectively detecting and responding to robbery events in real-time. Current approaches typically rely on singular object detection models, lacking the necessary accuracy and interoperability essential for robust security measures. This framework aims to mitigate the deficiencies of present surveillance systems by enabling accurate, efficient, and transparent detection of robbery events. Moreover, it seeks to facilitate swift communication with law enforcement agencies, thus enabling timely intervention in such situations.

It has the potential to significantly reduce crime rates, thereby fostering a safer environment for individuals and businesses. By employing advanced algorithms to identify potential threats in real-time, these projects enable law enforcement agencies to proactively address criminal activity, leading to a decrease in robbery incidents. Beyond the tangible economic benefits, robbery detection projects contribute to the overall well-being of communities by fostering a sense of security and trust among residents. Enhanced security resulting from robbery detection initiatives can boost business confidence and stimulate economic activity.

The project leverage advanced algorithms to identify patterns and anomalies indicative of criminal activity. By detecting and disrupting potential threats in real-time, these projects play a crucial role in preventing robberies and minimizing their impact on individuals and businesses. This proactive approach to crime prevention is essential in reducing overall crime rates and enhancing public safety.

II. Related Work

The authors of the paper [1] proposed a surveillance video framework in residential areas for detecting suspicious robbery activities using deep learning techniques. The authors utilized the YOLO algorithm for object detection and action identification. The authors aimed to reduce manual monitoring and provide real-time detection of robbery attempts in residential areas. The use of deep learning-based sensing fusion technique for action detection and recognition in continuous action streams was mentioned. The YOLO model was trained using the COCO dataset and modified to predict previously specified robberies based on the region of interest. The deep learning-based approach, particularly the CNN model, was employed for object detection, classification, and tracking in surveillance videos.

The authors of the paper [2] addressed the issue of improving the security of self-service banks by detecting robbery and violent scenarios. The authors extracted the motion region from the video and calculated the features such as length, width, energy, and orientation variance to distinguish robbery and violent segments from other videos.

The authors of the paper [3] conducted a systematic review of over 150 articles to explore the various machine learning and deep learning algorithms to predict crime. The research analysed the growth and popularity of the crime prediction domain, as well as the distribution of article page counts in the neighbourhood crime research. A word cloud was created to understand the key concepts or themes of the selected papers related to neighbourhood crime. The paper also described the methodology used for the literature search, including the identification of key terms and the construction of queries for different research databases

The authors of the paper [4] proposed a method for detecting suspicious objects from videos and analysing robbery events. The authors introduced a background subtraction technique using a minimum filter for detecting foreground objects. They also proposed a kernel-based tracking method for tracking moving objects and obtaining their trajectories. The paper presented ratio histogram for finding suspicious objects and accurately analysing their transferring conditions. Gaussian mixture models were used to model the visual properties of suspicious objects, allowing for accurate segmentation from videos. The authors of the paper [9] also discussed about using various models like two stream CNN and 3D CNN for action recognition. The authors of [10] used ResNet for the identification and further removal of raindrops from Images. The authors of paper [11] proposed a method for detecting faces in face masks. The author showed that faster-RCNN model has a high object detection rate and outperforming RCNN. The authors of [12] discussed about multi-model deep learning fusion in detecting stress in ECG. The author fused intermediate convolutional layers to produce better results.

III. Research Gap

1) Deficiency in Comprehensive Unified Frameworks: Despite the proliferation of studies addressing robbery detection using deep learning methodologies, a conspicuous gap emerges in the absence of comprehensive unified frameworks. While individual studies showcase the efficacy of various deep learning models, the integration of these disparate methodologies into a cohesive framework is conspicuously lacking. Consequently, there remains a dearth of research elucidating the development of an integrated solution that amalgamates the strengths of multiple advanced object detection models, including YOLOv8, Fast R-CNN, and RetinaNet. Such a unified framework could potentially offer heightened accuracy, robustness, and versatility in the detection of robbery events.

2) Inadequate Emphasis on Real-time Application Integration: Existing scholarship within the domain of deep learning-based robbery detection largely overlooks the exigency for seamless integration into real-time Android applications. While several studies showcase the efficacy of deep learning methodologies in detecting robbery events, there is a palpable scarcity of research dedicated to the practical integration of these techniques into operational surveillance systems. The absence of such research impedes the translation of theoretical advancements into tangible solutions capable of practical deployment in real-world security contexts.

3) Insufficiency in Explainable AI Validation Mechanisms: Despite the remarkable strides made in deep learning-based object detection and tracking, the pervasive opacity in the decision-making processes of these models poses significant challenges, particularly in critical applications such as security surveillance. A discernible research gap emerges in the scarcity of studies dedicated to the integration of explainable AI validation mechanisms within detection frameworks. Such mechanisms are imperative for fostering transparency, interoperability, and trustworthiness in the outputs generated by deep learning models, thereby mitigating concerns surrounding algorithmic biases and fostering stakeholder confidence in automated surveillance systems.

4) Scarcity of Exploration in Fusion Techniques: While extant literature showcases the efficacy of individual deep learning models for object detection and tracking, a conspicuous void exists in the exploration of fusion techniques aimed at amalgamating the strengths of multiple models. A dearth of research is observed in the investigation of fusion methodologies capable of collaboratively utilizing for enhanced effectiveness. the unique capabilities of YOLOv8, Fast R-CNN, and RetinaNet to enhance accuracy, robustness, and adaptability in detecting robbery events across diverse environmental contexts.

IV. Methodology

A. Dataset

The study utilizes the ARADS-9x800 dataset, focusing specifically on incidents of robbery. Comprising approximately 12,308 images categorized into six classes—namely, gun, knife, phone, robbery activity, and metal detector—the dataset also includes a partition into training, testing, and validation sets, with an 88% : 6% : 6% distribution. Each partition undergoes a series of augmentations, including the generation of three outputs per training example, as well as bounding box manipulations such as horizontal and vertical flips, minimum 0% and maximum 13% zoom cropping, exposure adjustments ranging from -15% to +15%, blur effects up to 1.75 pixels, and noise incorporation of up to 4% of pixels. This factor presents an advantageous aspect for training as it contributes to the enhanced realism and real-world applicability of the data. The dataset is formatted into two types, one for the YOLOv8 and other for the remaining two models, RetinaNet and Faster-RCNN. After augmentations, the train set has 25881 images and each of test and validation set has 1841 and 1840 images respectfully. Each of the models are trained, tested and validated with the augmented dataset for more robust feature training. For testing all the outputs, a edited 12 second video feed from news reports was made, which had sequences of robbery. All the models, including the fusion of these three were tested on it to find the frames of suspicious activity.

B. Faster RCNN

Faster-RCNN [6] is one of the three models that is utilized to detect robbery activity using the collected dataset. The input image undergoes a CNN backbone to extract feature maps, which capture various levels of visual information from the image. Specifically, Detectron2, Facebook's AI Research's next-generation library, provides state-of-the-art detection and segmentation algorithms. The Detectron2 model zoo includes the necessary modules to train the model. Within the detectron.data.datasets library, we utilize the register_coco_instances module to format and upload the dataset. The model is then trained with 4 workers and 4 images per training batch. A very low learning rate is selected to control parameter updates and prevent overshooting the optimal solution. The chosen number of iterations is 2000. Model evaluation is scheduled to occur every 500 iterations to monitor performance on the validation dataset, potentially enabling early stopping or adjustment of hyper-parameters based on validation metrics. The trained model is validated using a separate subset of the dataset to assess its generalization capabilities and ensure robust performance. **The Hyperparameters tuned for Faster-RCNN Model is given below**

```
cfg.DATALOADER.NUM_WORKERS = 4
cfg.MODEL.WEIGHTS = model_zoo.
get_checkpoint_url
("COCO-Detection/
faster_rcnn_X_101_32x8d_FPN_3x.yaml")
cfg.SOLVER.IMS_PER_BATCH = 4
cfg.SOLVER.BASE_LR = 0.0001
cfg.SOLVER.MAX_ITER = 2000
cfg.SOLVER.STEPS = [1000, 1500]
```

```
cfg.SOLVER.GAMMA = 0.05
cfg.MODEL.ROI_HEADS.
BATCH_SIZE_PER_IMAGE = 64
cfg.MODEL.ROI_HEADS.NUM_CLASSES = 6
cfg.TEST.EVAL_PERIOD = 500
os.makedirs(cfg.OUTPUT_DIR, exist_ok=True)
trainer = DefaultTrainer(cfg)
```

C. YOLOv8n

You Only Look At Once (YOLO) is a state-of-the-art, real-time object detection system. YOLOv8 [7] has an anchor-free architecture, which means it doesn't rely on anchor boxes, making it simpler to train with various datasets. It also includes an advanced backbone network and multi-scale prediction to enhance accuracy, especially for detecting small objects. YOLOv8 has shown improved accuracy compared to previous YOLO versions and competes with state-of-the-art object detection models [5]. The dataset is obtained and processed using the Roboflow platform, allowing for efficient management and augmentation of annotated images. The YOLOv8n model is trained on the custom dataset using the Ultralytics library. Hyperparameters such as epoch count set to 40, input image size set to 640, and confidence threshold are configured to optimize model performance. The trained model is validated using a separate subset of the dataset to assess its generalization capabilities and ensure robust performance. **The Hyperparameters tuned for YOLOv8n model is given below**

```
lr0: 0.01
# initial learning rate
lrf: 0.01
# final OneCycleLR learning rate
momentum: 0.937
# SGD momentum/Adam beta1
weight_decay: 0.0005
# optimizer weight decay 5e-4
warmup_epochs: 3.0
# warmup epochs
warmup_momentum: 0.8
# warmup initial momentum
warmup_bias_lr: 0.1
# warmup initial bias lr
box: 0.05
# box loss gain
cls: 0.5
# cls loss gain
cls_pw: 1.0
# cls BCELoss positive_weight
obj: 1.0
# obj loss gain
obj_pw: 1.0
# obj BCELoss positive_weight
iou_t: 0.20
# IoU training threshold
anchor_t: 4.0
# anchor-multiple threshold
# anchors: 3
# anchors per output layer
hsv_h: 0.015
# image HSV-Hue augmentation
hsv_s: 0.7
# image HSV-Saturation augmentation
hsv_v: 0.4
# image HSV-Value augmentation
translate: 0.1
# image translation
scale: 0.5
# image scale
fliplr: 0.5
# image flip left-right
mosaic: 1.0
# image mosaic
```

D. RetinaNet

RetinaNet [8] is a one-stage object detection model that utilizes a focal loss function to address class imbalance during training. Focal loss applies a modulating term to the cross entropy loss in order to focus learning on hard negative examples. In the implementation, Detectron2, known for its advanced detection and segmentation algorithms is deployed. By leveraging the Detectron2 model zoo, essential modules to train the RetinaNet model can be incorporated. Specifically, within the detectron.data.datasets library, we utilize the register_coco_instances module for dataset formatting and uploading, thereby streamlining the training process. During the training phase, the model is configured with with 4 workers and designate a training batch size of 4 images per batch. To enhance learning efficacy, a significantly low learning rate can be used to meticulously manage parameter updates to avoid overshooting the optimal solution. The training regimen spans 2000 iterations, offering abundant opportunities for the model to discern the underlying data patterns. Furthermore, to monitor the model's performance and ensure its generalization capability, model evaluation is scheduled for every 500 iterations, enabling early intervention such as halting training or adjusting hyper-parameters based on validation metrics. **The Hyperparameters tuned for RetinaNet Model is given below**

```
cfg.DATALOADER.NUM_WORKERS = 4
cfg.MODEL.WEIGHTS =
model_zoo.get_checkpoint_url
("COCO-Detection/
retinanet_R_101_FPN_3x.yaml")
cfg.SOLVER.IMS_PER_BATCH = 4
cfg.SOLVER.BASE_LR = 0.00025
cfg.SOLVER.MAX_ITER = 2000
cfg.SOLVER.STEPS = [1000, 1500]
cfg.SOLVER.GAMMA = 0.05
cfg.MODEL.ROI_HEADS.
BATCH_SIZE_PER_IMAGE = 64
cfg.MODEL.ROI_HEADS.NUM_CLASSES = 6
cfg.TEST.EVAL_PERIOD = 500
os.makedirs(cfg.OUTPUT_DIR, exist_ok=True)
trainer = DefaultTrainer(cfg)
```

E. Multi-Model Deep Learning Fusion

The fusion process lies at the core of the proposed methodology, wherein features extracted from the YOLOv8, RCNN, and RetinaNet models are amalgamated to construct a comprehensive representation of the input image. This amalgamation facilitates the synthesis of diverse and complementary information encoded within each model's feature space, ultimately enhancing the discriminate power and robustness of the fusion model in robbery detection tasks. The fusion process commences with the extraction of features from the pre-trained models, achieved through forward passes of input images through the respective model architectures. Specifically, the YOLOv8 model is instantiated to extract features capturing high-level semantic information regarding object presence and spatial localization. Concurrently, the RCNN model specializes in region-based feature extraction, delineating fine-grained details and contextual information within localized regions of interest. Likewise, the RetinaNet model excels in capturing multi-scale features, adeptly handling objects of varying sizes and aspect ratios. The below snippet combines the features of all the three models

```
combined_features = torch.cat
((features_yolov8,
features_rcnn['out'],
features_retinanet['out']), dim=1)
```

Subsequent to feature fusion, the fusion model applies additional layers, including convolutional and pooling operations, to further refine and distill the fused feature representation. These layers serve to extract higher-order features, enhance feature discriminate ability, and mitigate potential noise or redundancy within the feature space. The resulting refined feature representation embodies a comprehensive abstraction of the input image, poised to facilitate accurate and robust object detection. The below snippet specifies the configurations for the convolutional layer and the pooling layer.

```
extra_conv = nn.Conv2d
(256, 512, kernel_size=3, stride=1,
padding=1)
pooling = nn.MaxPool2d
(kernel_size=2, stride=2)
```

Now the new model is trained for 100 epochs with image size of 640 just like the other models.

F. Explainable AI

Explainable AI plays a crucial role in validation processes by providing detailed explanations for detected activities through the utilization of Layer-wise Relevance Propagation (LRP). This method enables the attribution of relevance scores to individual pixels within input images, thereby illuminating the specific regions that significantly influence the decision-making processes of the model. Post detection of suspicious activities, the frames are subjected to further analysis using LRP libraries such as PyTorch's Grad_cam, which contains essential modules for this purpose. Within Grad_cam, grad_cam.explain() class is designated to receive the model and class inputs, initiating an in-depth analysis of the output and subsequent explanation plotting. The snippet given below is used to plot the explanation given by the LRP.

```
lrp.plot_explanation
(frame=image,
explanation = explanation_lrp,
contrastive=False,
cmap='Reds')
```

G. Mobile Application

The mobile application serves as a comprehensive tool for facilitating both proactive and reactive responses to robbery events within its operational environment. Developed for Android version 7+ platforms, the application leverages a combination of sophisticated object detection models, namely YOLOv8, Fast R-CNN, and RetinaNet in addition to Explainable AI, which are seamlessly integrated into its architecture. These models are hosted on Firebase, the principal database housing critical application data including user profiles, incident reports, and the deployed detection models. Upon accessing the application, users are prompted to create an account, enabling personalized access to its functionalities. The application offers two primary features:

1) User-Initiated Incident Reporting: Through a dedicated reporting feature, application users can swiftly report instances of suspected robbery. Upon submission of such reports, the video files are converted into frames and the integrated detection models analyze the provided data, producing outcomes through an explainable AI module. In the event of a confirmed robbery detection, the application establishes a geo-fence around the user's location. Subsequently, pertinent data including latitude, longitude, and a video snippet capturing the detected event are transmitted to law enforcement agencies via cloud messaging. Additionally, users within the Geo-fenced area receive immediate alerts notifying them of the ongoing suspicious activity. The captured video snippet is securely stored within Firebase for future reference.

2) Live Camera Surveillance Integration: Integrating with live camera feeds, the application continuously monitors for potentially suspicious activities frame by frame. Upon detecting such events, the application dynamically establishes a Geo-fence around the affected area. Analogous to user-initiated reporting, relevant data comprising coordinates and stitched video footage are dispatched to law enforcement authorities via Firebase and cloud messaging channels. Furthermore, nearby application users are promptly notified of the detected activity. By encompassing both proactive reporting capabilities and live surveillance integration, the mobile application offers a multifaceted approach towards addressing robbery events. Its utilization of cutting-edge technologies not only enhances the efficacy of detection but also facilitates swift and informed responses, thereby contributing to the broader goal of enhancing public safety and security.

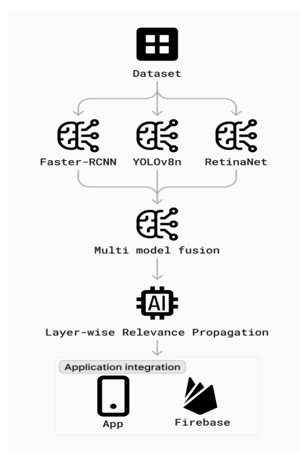


Fig. 1: Architecture diagram which gives an overview of Methodology

V. Performance Evaluation Metrics

- **Precision** and **Recall** are the main mathematical metrics which are used to assess the fused model. Precision metric assesses this model's ability to correctly classify a sample as positive. Recall metric assesses the model's ability to identify positive sample. Precision and recall are calculated for individual models and compared with fused model scores.

- Table I reveals that, the precision and recall of fused model is significantly higher than that of the individual models.
- It is evident that the efficacy of the fused model is high since the features of each of the models are fused to create a more robust model. The incorporation of convolutional and pooling layers in the fused model significantly contributes to enhancing its robustness. Figure 2 shows that the confidence score for the fused model is significantly higher compared to the individual models and consistent in all frames. This suggests the fused model is more certain about its suspicious activity detection. By combining the strengths of multiple models (e.g., YOLOv8, RetinaNet, Faster R-CNN), the fused model likely reduces overall uncertainty, leading to a more confident prediction.

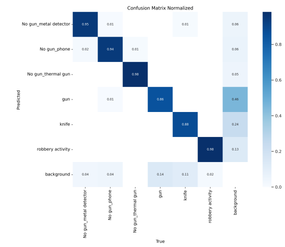
Fig. 2: Comparison between predictions from individual models and Multi-Model DL fusion

TABLE I: Comparison of Evaluation Metrics of each models and proposed Fusion Model

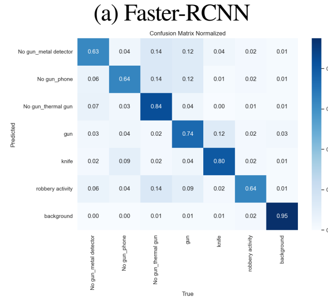
	Precision	Recall	F1 Score
Faster RCNN	0.726	0.683	0.70
YOLOv8n	0.784	0.729	0.76
RetinaNet	0.713	0.695	0.70
Proposed Fusion Model	0.951	0.911	0.93

- **Confusion Matrix** compares the actual labels of objects with the labels the model predicted. The confusion matrix of each of the individual models and the fusion model are calculated and compared. In a perfect confusion matrix, all the high values would be on the diagonal, which runs from the top left corner to the bottom right corner. This indicates that the model correctly classified each object.
- Each individual model has a disadvantage when we study the confusion matrix.
  - **Faster-RCNN** The confusion matrix suggests the model might struggle to differentiate between objects with similar appearances or that often co-occur in scenes. The model might occasionally mistake "background" for other objects and vice versa, as indicated by the off-diagonal values in the background row and column.
  - **YOLOv8n** The model struggles to differentiate between "gun" and "robbery activity". There is a 46% chance it might classify a gun as "robbery activity" and a 13% chance it might classify "robbery activity" as a gun. The model sometimes mistakes "background" for other objects. There's a 14% chance it might classify "background" as something else, and a 4% to 11% chance it might mistake other objects for "background".
  - **RetinaNet** The confusion matrix suggests that there is a confusion with the 'Background' class with 6% chance of confusion. The analysis suggests that while the model performs well in certain classes like 'gun' and 'No gun_metal detector', it struggles with distinguishing between 'knife' and 'robbery activity', potentially due to contextual features that are not necessarily indicative of a true relationship between these classes.

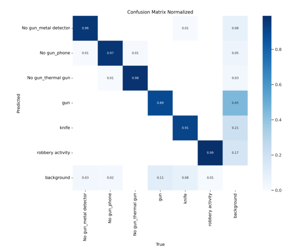
(a) Faster-RCNN



(b) YOLOv8



(c) RetinaNet



(d) Proposed Fusion Model

Fig. 3: Confusion Matrices for various models

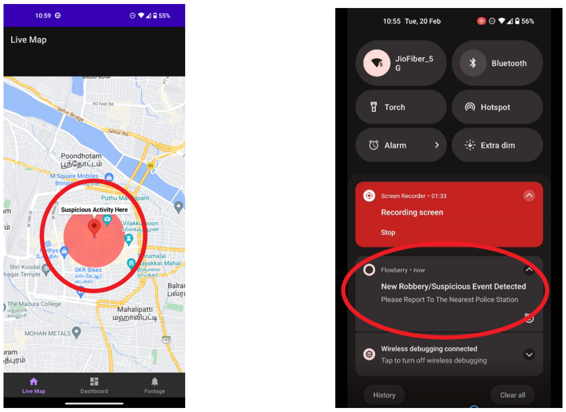
- Now, by analysing the confusion matrix calculated for the fused model, we can see that , the fused model has a lower confusion rate between these two classes, namely 'gun' and 'robbery activity' compared to what you might expect with the other three models. The fused model has higher values on the diagonal compared to the other three models, especially for "no_gun_metal_detector," "no_gun_thermal_gun," "knife," and "background". The fused model also shows a lower confusion rate between "gun" and "robbery activity" compared to the other models.
- Overall, the higher values on the diagonal and lower values off the diagonal in the fused model's confusion matrix suggest that it's performing better at correctly classifying objects compared to YOLOv8, RetinaNet, and Faster R-CNN on this particular dataset.

VI. Results

Upon model prediction of a potential robbery, the frames detected as robbery is sent to Explainable AI, which in turn plots the explanation for every frame which is detected as robbery. This will be used for validation of the suspicious activity. Push notifications are dispatched to app users situated within a predefined Geo-fence surrounding the event. These notifications contain precise event location details alongside recorded images, videos and the explanation plot stored in the back-end. Users are then directed to the footage section of the app, where they can assess the situation and determine the necessity for immediate action. Additionally, an SOS button is provided for emergency situations, enabling users to report to the nearest police station with collected evidence, subject to user consent. Furthermore, users can view the event location and assess the safety of their current whereabouts.

VII. Future Works

The integration of the Unified Detection Framework for Robbery Events, which encompasses YOLOv8, Fast R-CNN, and RetinaNet models, in conjunction with Explainable AI Validation and seamless integration into real-time Android applications, represents a notable advancement in the field of security surveillance. Nevertheless,



(a) The Geo-fenced area being marked when detected suspicious activity

(b) Application sending notification to users near the Geo-fence

Fig. 4: The application results after finding a suspicious activity

as is customary in scholarly pursuits, there are opportunities for additional exploration and refinement to augment the effectiveness and scalability of the proposed framework. Future research could focus on refining the fusion techniques employed to integrate multiple object detection models within the framework. Beyond the utilization of visual data exclusively from surveillance cameras, the integration of additional data modalities such as audio inputs, sensor data, and information sourced from social media platforms has the potential to significantly enhance the situational awareness within the framework and bolster the efficacy of robbery event detection.

VIII. Conclusion

In summary, the Unified Detection Framework for Robbery Events, which integrates YOLOv8, Fast R-CNN, and RetinaNet into a multi-model deep learning fusion model alongside Explainable AI Validation and real-time Android application integration, represents a substantial leap forward in the field of security surveillance systems. By amalgamating state-of-the-art object detection technologies and implementing explainable AI validation mechanisms, the framework showcases significant promise in augmenting the precision, effectiveness, and transparency of robbery event identification within practical contexts. The incorporation of this framework into an Android application not only amplifies its accessibility and user-friendliness but also empowers individuals to actively engage in incident reporting and fortify the overarching security infrastructure. By harnessing cloud-based storage and communication platforms, the framework facilitates smooth data interchange between users and law enforcement agencies, thereby enabling prompt and well-informed responses to identified incidents. In its essence, the Unified Detection Framework for Robbery Events serves as a testament to the profound impact of interdisciplinary collaboration and technological innovation in tackling current security challenges. Through ongoing refinement and expansion of its capabilities, we can endeavor to establish safer and more secure environments for communities on a global scale.

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